

Problem J

Gas Station

Time limit: 3 seconds

Problem Description

Alex is planning rest area placements on a simplified model of Taiwan's freeway system. The system contains n interchanges, connected by $n - 1$ bidirectional roads. The network is connected, and there is exactly one shortest route between any pair of interchanges. The i -th road connects interchanges u_i and v_i , and has a length of l_i .

Exactly k rest areas with gas stations can be built, each located at an interchange. A driver may start a trip from any interchange and travel to any other, always following the unique shortest path. They begin each trip with a full tank of gas and can refuel only at interchanges that have a rest area.

Alex is curious about the smallest possible fuel tank capacity d such that it's possible to place the k rest areas in a way that ensures no driver will ever run out of gas. On any trip, the driver must never have to travel more than d units along the path without passing through a rest area, including at the beginning or end of the journey. The goal is to figure out the minimum such d , assuming the rest areas are placed in the best possible way.

Input Format

The first line contains two integer n, k .

Followed by $n - 1$ lines, the i -th of which contains three integers u_i, v_i, l_i , representing the i -th road connects interchanges u_i and v_i with a length l_i .

Output Format

Output one integer, the smallest possible fuel tank capacity d .

Technical Specification

- $2 \leq n \leq 2 \times 10^5$
- $0 \leq k \leq n$
- $1 \leq u_i, v_i \leq n$
- $1 \leq l_i \leq 10^9$
- It is guaranteed that the input roads form a tree.

Sample Input 1

```
5 1
1 2 3
1 5 2
2 3 3
2 4 1
```

Sample Output 1

```
5
```

Sample Input 2

```
5 2
1 2 3
1 5 2
2 3 3
2 4 1
```

Sample Output 2

```
3
```